26-port L3 managed core routing switch

(NAV-C24S2Q)

High-Speed Layer 3 Industrial Connectivity

A high-performance L3 managed Ethernet switch oriented for next-generation IP metropolitan area networks, large campus networks, and enterprise networks.

Advanced Architecture

Equipped with ASIC switching chip and multi-core processor with 1.28Tbps switching capacity for high-density data centers.

Virtualization (VSS)

Virtualizes multiple physical devices into one logical device, extending clusters up to 80km and eliminating STP blocking.

High Reliability

Supports ISSU for uninterrupted upgrades, Ethernet OAM for fault detection, and redundant hot-swappable power supplies.

Switching Capacity

880Gbps

Non-blocking

Forwarding Rate

654 Mpps

@64byte

To meet the needs of network resource pooling in cloud computing data centers, the NAV-C24S2Q supports rich features such as BVSS virtualization. With data center core capabilities, it achieves access to more than 15,000 10G servers. It integrates multiple network services such as IPv6, network security, and traffic analysis, ensuring the network's longest uninterrupted communication capabilities.

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Product Overview

The **NAV-C24S2Q** is a high-performance Layer 3 (L3) managed Ethernet switch designed for next-generation IP metropolitan area networks, large campus networks, and enterprise environments. It features 24x 1/10G SFP+ fiber ports and 2x 40/100G QSFP28 fiber ports housed in a standard 1U/19-inch chassis.

Designed to meet the resource pooling demands of cloud computing data centers, the NAV-C24S2Q supports advanced virtualization features (such as BVSS). When deployed as a data center core switch, it scales to support access for over 15,000 10G servers, offering a comprehensive solution for ultra-large data centers.

Beyond high-performance L2/L3/L4 line-speed switching, it integrates critical network services including IPv6 support, network security, traffic analysis, and virtualization. It ensures maximum uptime through high-reliability technologies like continuous upgrades, graceful restarts, and redundant protection.

Key Features

Ultra-High Connectivity

- 24× 1/10G SFP+ fiber ports and 2× 40/100G QSFP28 uplinks for massive throughput.
- 1.28Tbps switching capacity with line-speed forwarding for high-density data centers.

VSS Virtualization

- Virtualizes multiple physical devices into one logical unit, extending clusters up to 80km.
- Bullet 2: Eliminates STP blocking to double performance and simplify management.

Rich L3 Service Features

- Full IPv4/IPv6 stack with static routing, RIP, OSPF, BGP, and IS-IS.
- Supports VRRP, BFD for OSPF, and ECMP for load balancing.
- Advanced multicast with IGMP v1/v2/v3 and PIM-SM/DM for HD video.
- 4K VLANs, QinQ, and private VLAN support for flexible segmentation.

Comprehensive Security

- IEEE 802.1x, Radius, and Tacacs+ authentication with hierarchical user permissions.
- Hardware defense against DoS, TCP/UDP floods, and broadcast storms.
- uRPF reverse routing lookup and deep packet inspection.

Carrier-Grade Reliability

- ISSU support for uninterrupted in-service software upgrades.
- Modular 1+1 redundant hot-swappable power supplies.
- Ethernet OAM (802.3ah) and BFD for millisecond-level fault detection

A next-gen core routing switch for cloud computing and large-scale campus networks, it features 100G uplinks and VSS virtualization, supporting access for over 15,000 servers with zero-downtime operations through advanced redundancy and hitless protection.

Technical Specification

Feature	Description
Model	NAV-C24S2Q
Fixed Port	24x 1/10G SFP+ fiber ports (Data) 2x 40G/100G QSFP28 fiber ports (Data)
Power Port	2 ports
Fan	4 (built-in)
Switching Capacity	880Gbps (non-blocking)
Forwarding Rate@64byte	654.72Mpps
MAC	32K
Buffer Memory	32M
Power Supply	AC100-240V 50Hz±10%
Operation Temp/Humidity	-10°C~+50°C, 5%~90% RH non-condensing
Storage Temp/Humidity	-20°C~+70°C, 5%~95% RH non-condensing
Dimension (L*W*H)	442.5*300*44.5mm
Datacenter Feature	Support VSS virtualization technology Configurable MAC address aging time Black hole MAC, MAC address learning quantity limit
MAC Exchange	Static configuration and dynamic learning of MAC addresses View and clear MAC addresses, MAC address filtering function
VLAN	GVRP, Private VLAN, 4K VLAN 1:1 and N:1 VLAN Mapping, Basic QinQ and QinQ functions
STP	802.1D (STP), 802.1W (RSTP), 802.1S (MSTP) BPDU protection, Root protection, Loop protection Multicast traffic cross-VLAN replication
Multicast	Support multicast group policy and multicast group quantity limit IGMP v1/v2/v3, PIM-SM, PIM-DM, IGMP Snooping, IGMP Fast Leave
IPv4	Policy routing, BFD for OSPF, BGP Equal-cost routing to achieve load balancing Static routing, RIP v1/v2, OSPF, BGP, IS-IS, BEIGRP
IPv6	MLD v1/v2, MLD Snooping ICMPv6, DHCPv6, ACLv6, IPv6 Telnet Manual tunnel, ISATAP tunnel, 6to4 tunnel IPv6 static routing, RIPng, OSPFv3, BGP4+ IPv6 neighbor discovery, Path MTU discovery

QoS	802.1P/DSCP priority remarking CAR traffic limit, Traffic policing and traffic shaping Queue scheduling methods such as SP, WRR, SP+WRR, etc. Congestion avoidance mechanisms such as Tail-Drop and WRED Traffic classification based on each field of the L2/L3/L4 protocol header
Security	Ingress and Egress ACL, match L2/L3/L4 and IP quintuple, copy, forward and discard Hash origin and same destination load balancing to ensure session integrity of traffic output URPF, IEEE 802.1x certified, Radius, Tacacs+ certified IP+MAC+port binding, DHCP Snooping, DHCP Option 82 Port security, Port isolation, command line hierarchical protection Suppression of multicast, broadcast, and unknown unicast packets Prevent DDoS attacks, TCP SYN Flood attacks, UDP Flood attacks ACL flow identification and filtering security mechanism based on L2/L3/L4
Reliability	Optional power supply 1+1 backup GR for OSPF, BGP, BFD for OSPF, BGP VRRP hot standby protocol, EAPS, ERPS ring network protection Static/LACP link aggregation, support for cross-service card link aggregation
Management	SNMP v1/v2/v3, Browser-based Web management ZTP (Zero Touch Provisioning), RMON event history Console, Telnet, SSH 2.0, TFTP file upload and download management
Green Energy Saving	IEEE 802.3az green energy-efficient Ethernet

Ordering Information

Model	Description	Power Supply
NAV-C24S2Q	L3 managed Ethernet core routing switch with 24x 1/10G SFP+ fiber ports and 2x 40/100G QSFP28 fiber ports. Redundant dual AC power supply (75W*2). Supports 1U/19" cabinet mount.	75W x 2
Note: Optical modules are not included and must be purchased separately.		

Packing List

Content	Quantity	Unit
NAV-C24S2Q (26-port 10G Switch)	1	Set
AC Power Cable	2	PC
RJ45-DB9 Adapter Cable	1	PC
Mounting Ear	1	Set
User Guide	1	PC
Warranty Card & Certificate	1	PC

Optical Module

Product	Model	Description	Unit
1.25G Optical Module	2630	SFP optical module, 1.25G multi-mode dual fiber 850nm, transmission distance: 550m, LC interface. Supports DDM function and hot plugging.	PC
	2632	SFP optical module, 1.25G single-mode dual fiber 1310nm, transmission distance: 20km, LC interface. Supports DDM function and hot plugging.	PC
	2612-T	SFP optical module, 1.25G single-mode single fiber TX1310nm/ RX1550nm, transmission distance: 20km, LC interface. Supports DDM function and hot plugging.	PC
	2613-R	SFP optical module, 1.25G single-mode single fiber TX1550nm/ RX1310nm, transmission distance: 20km, LC interface. Supports DDM function and hot plugging.	PC
	2612-T-SC	SFP optical module, 1.25G single-mode single fiber TX1310nm/ RX1550nm, transmission distance: 20km, SC interface. Supports DDM function and hot plugging.	PC

	2613-R-SC	SFP optical module, 1.25G single-mode single fiber TX1550nm/ RX1310nm, transmission distance: 20km, SC interface. Supports DDM function and hot plugging.	PC
Power Module	2633	1.25G SFP optical module transfers to 10/100/1000M RJ45 port.	PC
10G Optical Module	6630	SFP+ optical module, 10G multi-mode dual fiber 850nm, transmission distance: 300m, LC interface. Supports DDM function and hot plugging.	PC
	7832	SFP+ optical module, 10G single-mode dual fiber 1310nm, transmission distance: 20km, LC interface. Supports DDM function and hot plugging.	PC
	7832-33	SFP+ optical module, 10G single-mode single fiber TX1330nm/ RX1270nm, transmission distance: 20km, LC interface. Supports DDM function and hot plugging.	PC
	7832-27	SFP+ optical module, 10G single-mode single fiber TX1270nm/ RX1330nm, transmission distance: 20km, LC interface. Supports DDM function and hot plugging.	PC